LUT Business School

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Modelling and Forecasting Monthly Stock Returns of Neste Oyj Using the ARMA Model.

Research Plan

Declaration of AI Usage: I hereby state that no any kind of AI tool was used for any purpose during this assignment."

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1. Introduction

This research aims to predict the monthly return of Neste Oyj, a Finnish company focused on the production of sustainable aviation fuel (SAF) and renewable diesel. Neste's goal is to support the reduction of greenhouse gas emissions in both the transportation and industrial sectors. Originally operating as Finland's state petrol company, the company made its first appearance on the Nasdaq Helsinki (formerly known as OMX Helsinki) in April 2005. (Neste 2025).

To achieve the research objective of this paper, obtaining time series data is a prerequisite. With a long trading history of over 20 years, Neste Oyj fulfils this prerequisite as it possesses a sufficiently large and reliable dataset for this econometric modelling task. Aside from this ample availability of time series data, the stock also exhibits other characteristics suitable for ARMA Modelling, such as operating in different sectors: energy, renewables, and refining, which are sensitive to commodity price movements, global demand shifts, and regulatory changes. These factors may create autocorrelation (key characteristics of ARMA Models) and short-term dependencies, which are what ARMA models are meant to capture.

In addition to the aforementioned, stock returns are good illustrators of broader macroeconomic conditions, indicating that predicting Neste's stock performance could not only give investors insights about the company but also understand the Finnish economy's response to global market dynamics.

2. Data Collection

The ideal place to get data for such an econometrics analysis will be from Yahoo Finance or Nasdaq Nordic, as these platforms provide a complete, clean, comprehensive, and downloadable dataset of daily, monthly, and yearly stock prices in euros in different formats, one being CSV good for voluminous data. (Nasdaq Nordic, 2024; Yahoo Finance, 2024). In this case, the research needs to analyse the stock's historical closing prices for the past five years (January 2020 - December 2024). It is relevant and advisable to collect such data from these official and publicly accessible databases, as they are accurate and transparent.

After getting the raw data, the study's next focus will be on converting the adjusted closing price to returns. Adjusted prices are preferred because they account for corporate actions such as dividends and stock splits, thus providing a consistent measure of stock performance over time (Brooks, 2019). From these prices, monthly log returns are calculated using the standard transformation $r_t=Ln(P_t/P_{t-1})$, which is widely used in financial econometrics to stabilise variance and help achieve stationarity in the series (Tsay, 2010). The result of this operation is the main variable (univariate) for modelling and forecasting within the ARMA framework.

3. Methodology Assessment

The Auto Regressive Moving Average ARMA univariate framework is the selected model for this research. This is because the data is likely to possess short-term autocorrelation and weak stationarity, features commonly observed in stock market time series (Brooks, 2014; Tsay, 2010).

Also, given that the research solely aims at predicting mean of the stock returns rather than volatility of univariate structural operation, ARMA is a convenient option The ARMA model consists of two components: the autoregressive (AR) part, which captures the influence of past returns on current returns, and the moving average (MA) part, which reflects the effect of past random shocks (Jothimani et al., 2015). Also, a couple of studies have demonstrated the strength and relevance of ARMA in solving same or similar topics related to this research. For instance, Jothimani, Shankar, and Yadav (2015) demonstrated that ARMA models effectively captured short-run dependencies in stock indices, producing accurate forecasts across various market conditions. Additionally, Rounaghi and Nassir Zadeh (2016) found that ARMA models produced reliable forecasts of monthly and yearly returns in major markets such as the S&P 500 and the London Stock Exchange. Another recent study is that of Rahman, Uddin, and Reza in 2023, which verifies how ARMA models continue to perform competitively in predicting stock price settings, especially when return series display limited noise and little autocorrelation.

Other models, such as GARCH-type models (GARCH and EGARCH), were considered for this research, but GARCH is designed primarily for forecasting conditional variance or volatility rather than the expected return series (Rounaghi & Nassir Zadeh, 2016). Similarly, Vector Auto Regressive (VAR) models could also be an alternative, but they suit the analysis of dynamic relationships among multiple time series, such as between a stock and a market index; they add unnecessary complexity when analysing a single stock (Rahman et al., 2023).

To fully implement the ARMA (p, q) Model, the Box–Jenkins methodology will be applied. This methodology involves assessing the stationarity of the variable, determining optimal lag orders using Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots through Akaike (AIC) and Bayesian (BIC) information criteria, estimating parameters, and verifying model adequacy through the Ljung–Box test for residual autocorrelation.

4. Possible Limitations

As with every model, there are risks of methodological and interpretational issues when using the ARMA model on such data. An issue could be that of non-stationarity, which could bias parameter estimates and invalidate model assumptions (Tsay, 2010). Moreover, financial time series data, such as return in this case, could show structural breaks originating from market shocks or policy changes that tend to affect parameter stability (Rounaghi & Nassir Zadeh, 2016). To prevent these challenges, the Augmented Dickey–Fuller (ADF) test is used to confirm stationarity, and mathematical operations such as differencing or logarithmic operation on returns will stabilise the series (Brooks, 2014). If structural instability is observed, the model can be re-estimated for sub-periods or increased with dummy variables to account for regime changes.

During the modelling phase, heteroskedasticity could be encountered, meaning that the stock return volatility often changes over time, and this becomes hard for the ARMA model to capture (Rahman, Uddin & Reza, 2023). An ARMA–GARCH extension could be considered to improve accuracy; in addition to the aforementioned, the residuals show volatility clustering. These measures will ensure the interpretability of the forecasting results.

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